

DropBox

functional requirements:

user should be able to upload/download any file based on his preference,

user should be able to search for any file based on his preference,

Non-functional requirements:

data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine we can prioritise 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

user,

file,

history,

API :

POST:/upload

{

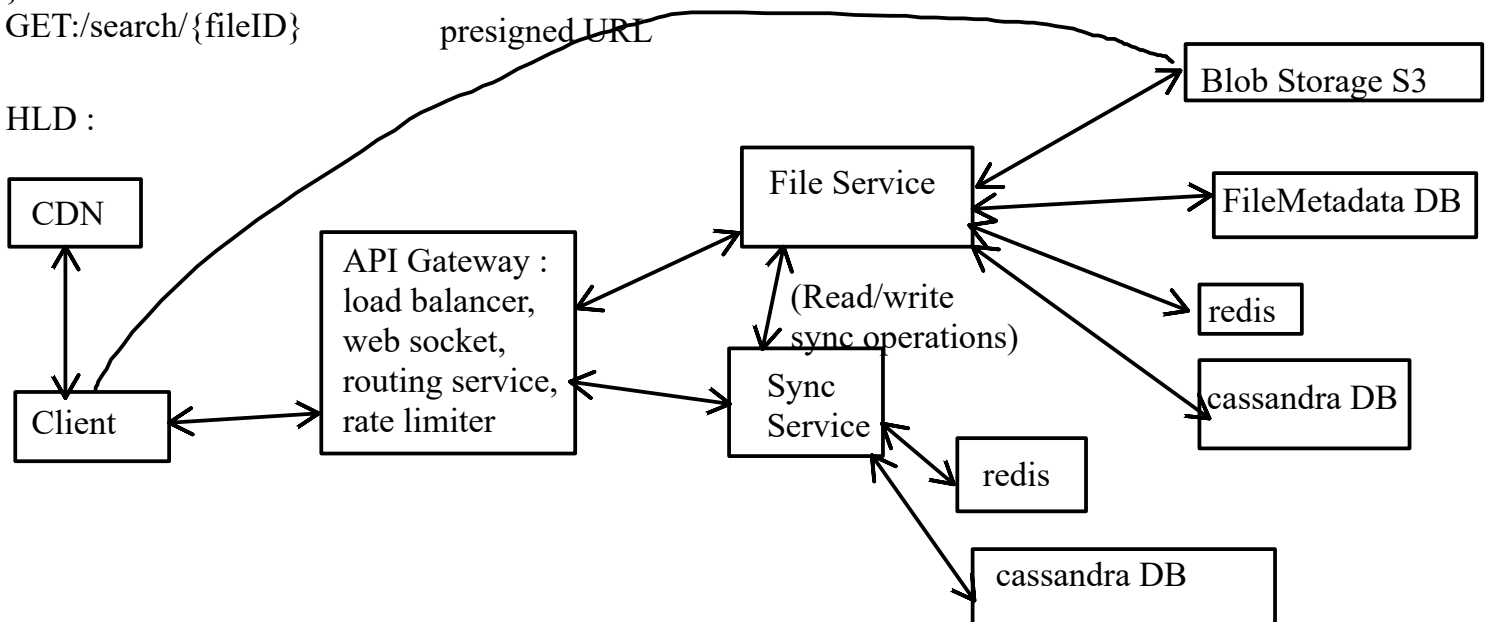
userID,

filemetadata

}

GET:/search/{fileID}

HLD :



- 1) we need to chunk 50GB file into multiple chunks
- 2) fingerprint/hash each of these chunks separately
- 3)upload each piece of these chunks separately
- 4) update completed status for each of these chunks separately